

Project Executable Files

Date	06 July 2026
Team ID	Not applicable / not assigned
Project Name	Credit Card Approval Prediction
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for the project. All files are organized as described below and uploaded to the submission platform (GitHub repository).

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	No
3	requirements.txt / package.json / dependency file	No
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	No
10	Demo video or walkthrough (if required)	No

Step 2: File / Folder Structure

```
Credit-Card-Approval-Prediction/  
  application_record.csv - Raw applicant demographic/financial data  
  credit_record.csv      - Monthly credit/payment status records  
  main.py                - Data loading, preprocessing, EDA, model training & comparison  
                        (Logistic Regression, Decision Tree, Random Forest)  
  model.pkl              - Serialized trained Random Forest model  
  app.py                 - Flask web application (serves prediction form & results)  
  templates/             - HTML pages for the Flask app (input form, result page)  
  static/                - CSS / static assets for the web UI  
  .venv/                 - Local Python virtual environment (not required by evaluator)
```

Note: No README.md or requirements.txt is currently present in the repository. Required third-party packages (installed manually) are: pandas, numpy, scikit-learn, flask.

Step 3: Deployment / Access Details

Field	Details
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Hosted / Deployed URL	Not yet deployed (deployment planned on Render)
Login Credentials (Demo)	Not applicable - no authentication/login required
Platform / Hosting Provider	Render
Repository Link	https://github.com/JoyalKumar1407/Credit-Card-Approval-Prediction
Demo Video Link	Not applicable

Step 4: Run Instructions

Pre-requisites:

Python 3.8+, pip. Required packages: pandas, numpy, scikit-learn, flask.

Step 1: Clone the repository:

```
git clone https://github.com/JoyalKumar1407/Credit-Card-Approval-Prediction.git
```

Step 2: Move into the project folder:

```
cd Credit-Card-Approval-Prediction
```

Step 3: Install dependencies (no requirements.txt is provided, so install manually):

```
pip install pandas numpy scikit-learn flask
```

Step 4: (Optional) Retrain the model - this regenerates model.pkl from the CSV datasets:

```
python main.py
```

Step 5: Start the Flask application:

```
python app.py
```

Step 6: Open a browser and go to:

```
http://127.0.0.1:5000
```

Step 7: Fill in the applicant details in the form and submit to view the prediction (Good Customer / Bad Customer).

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	No requirements.txt or README included in the repository	Install dependencies manually: pandas, numpy, scikit-learn, flask
2	Categorical fields are label-encoded during training but the encoder mappings are not saved separately, so form inputs in app.py must match training-time encoding order	Values in the web form are kept consistent with the training script's category order; a saved encoder/pipeline is a planned improvement
3	Flask app currently runs with debug=True, and no requirements.txt/Dockerfile is provided for deployment	Disable debug mode and add requirements.txt before deploying to Render